METRIC

MIL-PRF-85045/21(SH)
W/AMENDMENT 1
18 April 2017

SUPERSEDING
MIL-PRF-85045/21
26 July 1999

PERFORMANCE SPECIFICATION SHEET

CABLE, FIBER OPTIC, EIGHT FIBERS, ENHANCED PERFORMANCE, CABLE CONFIGURATION TYPE 2 (OFCC), FOR SUBMARINE OUTBOARD USE ONLY (NOT FOR INBOARD USE), CABLE CLASS SM AND MM, (METRIC)

This specification is approved for use by Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-85045.

CLASSIFICATION:

Fiber optic cable configuration type: 2 (OFCC).

Fiber Cable Class: MM (Graded-index, glass core and glass cladding, multimode).

SM (Dispersion-unshifted, glass core and glass cladding, single-mode).

DESIGN AND CONSTRUCTION:

Fiber:

Class MM fibers shall be in accordance with MIL-PRF-49291/6.

Class SM fibers shall be in accordance with MIL-PRF-49291/7.

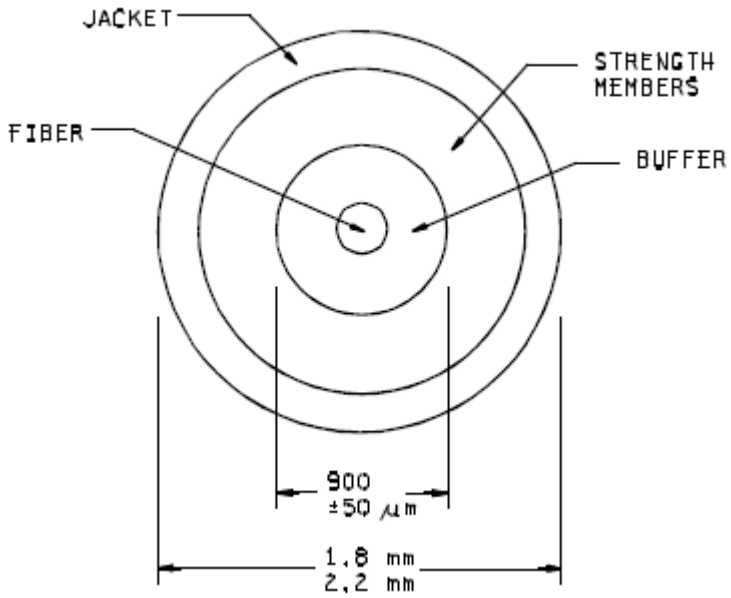
Buffer diameter: 900±50 µm.

OFCC:

Dimensions and configuration: See figure 1.

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NOTE:

1. Dimensions are in millimeters.

FIGURE 1. Optical fiber cable component.

Mass per unit length: ≤ 15 kg/km.

Tensile loading: ≥ 270 N.

Dynamic bend tensile load: 90 N minimum.

Jacket material: The OFCC jacket shall be composed of a low halogen, low smoke, and low toxicity polymer material.

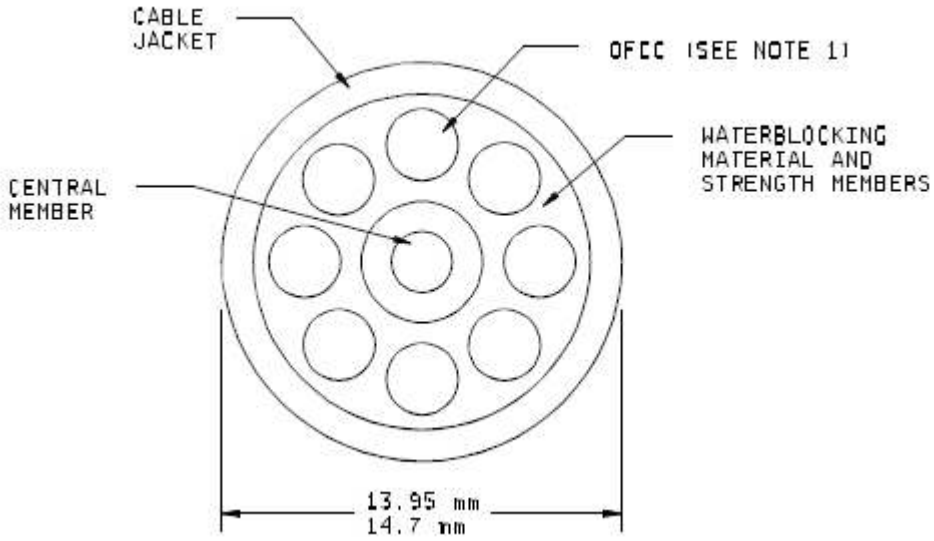
Short term minimum bend diameter: Eight times the OFCC outer diameter.

Long term minimum bend diameter: Sixteen times the OFCC outer diameter.

Finished cable:

Dimensions and configuration: See figure 2. Eight OFCC units shall be helically laid over the central member. Materials with acoustic performance properties similar to those of Mylar shall not be used. The outer jacket may be composed of multiple layers. The total jacket wall thickness shall be a minimum of 2.8 mm (establishes compatibility with termination and penetration devices).

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NOTE:

1. OFCC – Optical fiber cable component.

FIGURE 2. Eight OFCC fiber optic cable.

Number of fibers: 8 (one per OFCC).

Concentricity: ≥ 0.65.

Mass per unit length: ≤ 200.0 kg/km.

Jacket material: The outer jacket shall be arctic type polychloroprene (black).

Short term minimum bend diameter: Eight times the cable outer diameter. (The short term minimum bend diameter is to be used in all environmental and mechanical tests which specify a cable minimum bend diameter.)

Long term minimum bend diameter: Sixteen times the cable outer diameter.

Minimum continuous length: The minimum continuous length of all cables shall be not less than 0.5 km. If lengths less than 0.5 km are specified in the purchase order, Quality Conformance Inspection shall be performed on test specimens not less than 0.5 km in length from which the purchase order lengths are cut.

PERFORMANCE REQUIREMENTS:

Optical Properties:

Maximum attenuation rate: 4.5 dB/km at 850±20 nm, 2.0 dB/km at 1300±20 nm for class MM fiber.

2.0 dB/km at 1310±20 nm and 1550±20 nm for class SM fiber.

Bandwidth: Fiber with a minimum bandwidth of 500 MHz-km at 1300 nm and 160 MHz-km at 850 nm shall be used (multimode cables only).

Change in optical transmittance: Measurements to be made at 1300±20 nm.

Crosstalk: Applicable.

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Mechanical properties:

Tensile loading and elongation: Applicable, tensile loading ≥ 2700 N.

Operating tensile loading: Applicable.

Low temperature flexibility (cold bend): The exposure temperature shall be -40 °C.

Cyclic flexing: 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. Change in optical transmittance measurements are to be made every 100 cycles for the 500 cycle exposure and every 25 cycles for the 100 cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same test position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Crush: Applicable.

Cable twist bending: 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. Change in optical transmittance measurements are to be made every 100 cycles for the 500 cycle exposure and every 25 cycles for the 100 cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same test position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Radial compression: Applicable.

Impact: 50 cycles at 25±2 °C and 20 cycles at -40±2 °C. Upon final visual examination at all tested temperatures, there shall be no jacket damage such as splitting or cracking.

Pressure cycling: Applicable. The change in optical transmittance shall be less than 1.0 dB during the test and 0.5 dB after the test.

Hosing: Both low pressure and hydrostatic pressure are applicable.

Low pressure test pressure shall be 379 kPa. Leakage shall be not greater than 400 ml after 2 hours and not greater than 1000 ml after 6 hours.

Hydrostatic: 7.7 MPa. There shall be no leakage through the cable core during the first two hours. The total leakage (cable core and OFCC) shall be not greater than 3000 ml after 6 hours.

Dripping: Applicable.

Cable scraping resistance: 750 cycles.

Cable to cable abrasion: 500 cycles.

Environmental properties:

Temperature range:

Operating: -28 to +65 °C.

Nonoperating: -40 to +70 °C.

Storage: -40 to +70 °C.

Temperature cycling: Change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made over a period of 1 hour at the end of each temperature plateau.

Temperature/humidity cycling: Change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made at the end of each temperature plateau.

Storage temperature: Applicable.

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Life aging: The jacket material shall be tested at 100 °C for 168 hours. Tensile strength and elongation retention shall be not less than 50%.

Freezing water immersion (ice crush): Applicable.

Weathering: Applicable.

Fluid immersion: Exposure to automobile gasoline and tap water are not required and the following test temperatures shall be used for the fluids indicated: fuel oil (98 to 100 °C), turbine fuel (48 to 50 °C) and lubricating oil (98 to 100 °C).

Shock: Applicable, external wireway mounting shall be used.

Water absorption: 10 mg per square centimeter maximum.

Paint susceptibility: Applicable.

Chemical properties:

Halogen content: <0.2 percent

Cross-link verification: This test is applicable for the cable jacket only. The test shall be conducted in accordance with ICEA standard T-28-562 and run at 200 °C. The test shall be sequenced after the weathering test in the qualification test sequence and after the fluid immersion test in the group C quality conformance test sequences. The hot creep shall not exceed 100 percent and the hot creep set shall not exceed 10 percent.

Part or identifying number (PIN):

M85045/21-01 (Multimode).

M85045/21-02 (Single-mode).

Qualification and conformance inspection:

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/ 2/ 18/	Conformance inspection
I	Visual and mechanical inspection	3.4, 3.9, 3.10	4.7.2	3 samples, 0.5 km each 3/	A
	Attenuation rate	3.5.1	4.7.4.1	3 samples, 0.5 km each 4/	A
II	Crosstalk	3.5.3	4.7.4.3	3 samples, 0.5 km each 4/	
III	Temperature cycling	3.7.1	4.7.6.1	2 samples, 0.5 km each 4/ (1 on reel, 1 off reel)	C
	Temperature humidity cycling	3.7.3	4.7.6.3	2 samples, 0.5 km each 5/	C
	Storage temperature	3.7.4	4.7.6.4	2 samples, 0.5 km each 5/	
	Low temperature flexibility (cold bend)	3.6.1	4.7.5.3	2 specimens, 8 m each 6/	B
	Cyclic flexing	3.6.4	4.7.5.4	6 specimens, 5 m each 7/ (2 specimens at each temperature)	
	Crush	3.6.5	4.7.5.5	2 specimens, 5 m each 6/	
	Cable twist-bending	3.6.6	4.7.5.6	6 specimens, 5 m each 7/ (2 specimens for each temperature)	C

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Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/ 2/ 18/	Conformance inspection
	Impact	3.6.8	4.7.5.8	2 specimens, 5 m each 6/	C
	Temperature life (life aging)	3.7.6	4.7.6.6	2 specimens, 300 m each 6/	C
	Tensile loading and elongation	3.6.1	4.7.5.1	2 specimens, 150 m each 8/	C
	Operating tensile load	3.6.1.1	4.7.5.1.1	2 specimens, 150 m each 9/	C
	Freezing water immersion (ice crush)	3.7.7	4.7.6.7	2 specimens, 30 m each 6/	
	Fungus resistance	3.8.4	4.8.4	2 specimens, 0.5 m each 6/	
	Cable element removability	3.6.18	4.7.5.18	2 specimens, 0.5 m each 6/	C
IV	Thermal shock	3.7.2	4.7.6.2	1 specimen, 0.49 km 4/ (on reel)	
	Weathering	3.7.8	4.7.6.8	1 specimen, 2 m 10/ and 3 material samples 11/	
	Fluid immersion	3.7.9	4.7.6.9	1 specimen, 2 m 10/ and 3 material samples 11/ for each specified fluid	C
	Paint susceptibility	3.7.15	4.7.6.15	2 specimens, 2 m 12/	
	Jacket self-adhesion or blocking	3.7.11	4.7.6.11	1 specimen, 30 m 10/	
	Shock	3.7.13	4.7.6.13	1 specimen, 30 m 10/	
	Dynamic bend	3.6.2	4.7.5.2	1 specimen, 150 m 13/	B
	Hosing: Low pressure	3.6.12.1	4.7.5.12.1	1 specimen, 1.5 m 10/	C
	Hosing: Hydrostatic	3.6.12.2	4.7.5.12.2	1 specimen, 2 m 10/	C
	Radial compression (for application B only)	3.6.7	4.7.5.7	1 specimen, 10 m 10/	
	Pressure cycling (for application B only)	3.6.11	4.7.5.11	1 specimen, 30 m 10/	C
	Corner bend	3.6.9	4.7.5.9	2 specimens, 5 m 12/	
	OFCC kink	3.4.2.2.2	4.7.3.5	2 specimens, 1 m 10/	
V	Dripping	3.6.13	4.7.5.13	1 specimen, 30 cm 10/	
	Cable jacket tear strength	3.6.14	4.7.5.14	3 specimens, 1 m each 12/	C
	Cable jacket material tensile strength and elongation	3.6.15	4.7.5.15	5 specimens 14/	
	Cable abrasion resistance	3.6.16	4.7.5.16	4 specimens, 2 m each 15/	C
	Cable shrinkage	3.6.17	4.7.5.17	3 specimens, 0.5 m each 12/	C

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Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/ 2/ 18/	Conformance inspection
	Durability of identification marking	3.6.19	4.7.5.19	3 specimens, 2 m each 12/	
	Water absorption	3.7.14	4.7.6.14	2 specimens 16/	
VI	Acid gas generation (OFCC only)	3.8.1	4.8.1	1 specimen, 1 m 17/	C
	Halogen content (OFCC only)	3.8.2	4.8.2	1 specimen, 1 m 17/	
	Toxicity index (OFCC only)	3.8.3	4.8.3	1 specimen, 1 m 17/	C
	Smoke index (OFCC only)	3.8.5	4.8.6	1 specimen, 1 m 17/	C

NOTES:

1/ Tolerance on 0.5 kilometer length is ±10 percent provided results are normalized to 1 kilometer.
2/ Tolerance on shorter lengths is ±10 percent.
3/ The visual and mechanical inspection shall only be conducted on a 2-meter section of each sample.
4/ The same samples as used in the visual and mechanical inspection shall be used.
5/ The same samples as used in the temperature cycling test shall be used: 1 on reel, 1 off reel.
6/ A specimen cut from each sample used in the temperature cycling test shall be used.
7/ Three specimens cut from each sample used in the temperature cycling test shall be used.
8/ A specimen cut from each sample used in the life test shall be used.
9/ The same sample as used in the tensile loading and elongation test shall be used.
10/ A specimen cut from the specimen used in the thermal shock test shall be used.
11/ Three jacket material samples as specified in 4.7.5.15 shall be used in this test.
12/ Specimens cut from the specimen used in the thermal shock test shall be used.
13/ A specimen cut from the specimen used in the thermal shock test or one of the specimen used in the tensile loading and elongation test shall be used.
14/ Jacket material samples as specified in 4.7.5.15 shall be used in this test.
15/ Specimens cut from the specimen used in the thermal shock test shall be used. Two specimens shall be used for scraping abrasion testing and two specimens shall be used in the cable-to-cable abrasion testing.
16/ Jacket material samples as specified in 4.7.5.15 shall be used.
17/ A specimen of OFCC cut from one of the samples used in the temperature cycling or thermal shock tests shall be used.
18/ Mixed fiber cables, consisting of alternating single-mode and multimode fibers, can be used for qualification purposes. If a failure occurs for any fiber in a mixed fiber cable, then all of the fibers in the cable are considered to have failed (both single-mode and multimode).

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Qualification by similarity. DELETED

Manufacturers who are qualified under this specification sheet for multimode (single-mode) fiber cable and whose single-mode (multimode) fiber cable passes the visual and mechanical, attenuation rate, temperature cycling, temperature/humidity cycling, storage temperature, cyclic flexing, crush, cable twist-bending, impact (low temperature only), tensile loading and elongation, operating tensile loading, thermal shock and dynamic bend inspections specified herein, are qualified under this specification sheet for single-mode (multimode) fiber cable. This qualification by similarity is applicable if the only difference between the previously qualified cable and the cable under test is that the optical fiber has been changed from a multimode (single-mode) fiber to a single-mode (multimode) fiber. Testing may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order must be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

Group	Qualification inspection	Qualification inspection 1/	Conformance inspection 2/
I	Visual and mechanical inspection	X	A
	Attenuation rate	X	A
II	Temperature cycling	X	C
	Temperature humidity cycling	X	C
	Storage temperature	X	
III	Cyclic flexing	X	
	Crush	X	
	Cable twist-bending	X	C
	Impact (low only)	X	C
	Tensile loading and elongation	X	C
	Operating tensile load	X	C
IV	Thermal shock	X	
	Dynamic bend	X	B

NOTES:

1/ Where manufacturers are qualified for M85045/21-01 and are pursuing qualification for M85045/21-02.

2/ Where manufacturers are qualified for M85045/21-01 and M85045/21-02 and are performing conformance testing for M85045/21-02.

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Custodian:
Navy – SH

Review activity:
DLA – CC

Preparing activity:
Navy – SH
(Project 6015-2015-001)

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